From Structure to Story

Navigating an International Career in Artificial Intelligence

or, A Researcher's Tale: There and Back Again

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A Word of Warning

This talk is only partly about my research.

It is mainly about what I have learned as an international researcher.

It may get personal at times.

And it is full of Tolkien references, for which I apologize.

I don't know half of you half as well as I should like. — Bilbo Baggins

Concerning Olga

It's a dangerous business, going out your door.
— Bilbo Baggins



Where It Happened



Akademgorodok

where I grew up



Moscow State University

the humanities



University of Washington

computer science, then computational linguistics

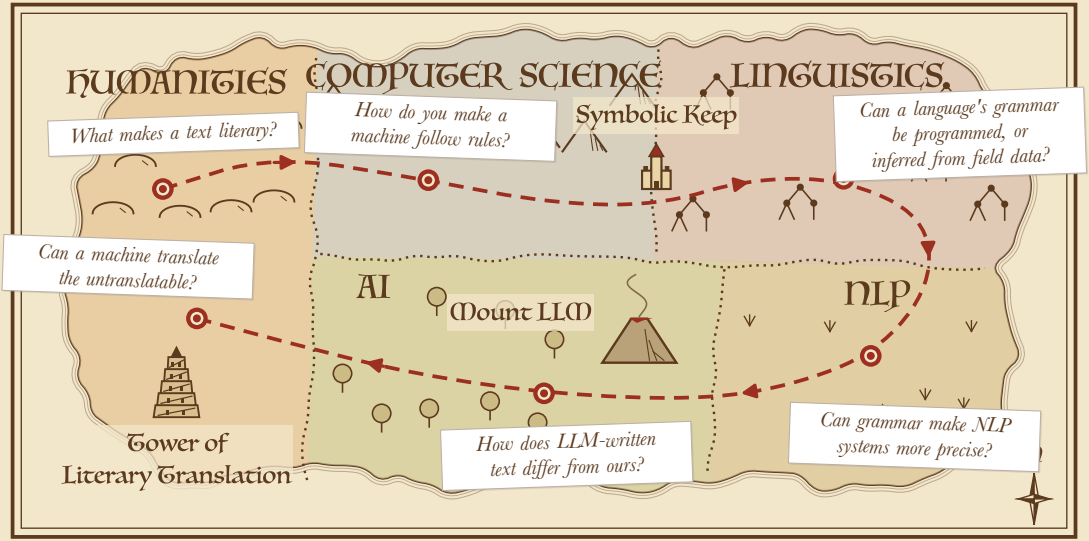


A Coruña

a postdoc, and a new trajectory

Here Be Disciplines

Not all those who wander are lost.
— J.R.R. Tolkien



Testing Multilingual Syntactic Theory

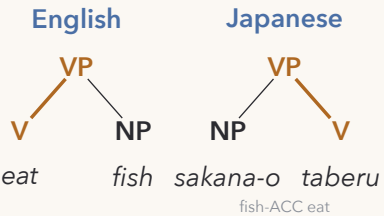


The framework is tested on languages from all over the world. In red: the ones I worked on. One principle has to hold across all of them.



One principle, opposite orders

a phrase inherits its head's category



Zamaraeva et al. (2022), *Journal of Language Modelling*; coordinates from WALS.

A Keep Built by Many



The Grammar Matrix: two decades of work by Emily M. Bender, her students and collaborators. My stones are in amber.

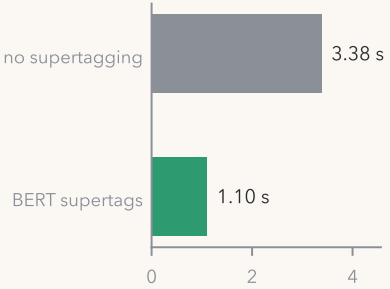
Clausal modifiers Howell & Zamaraeva 2018	**Clausal complements** Zamaraeva, Howell & Bender 2019	**Constituent questions** Zamaraeva 2021	
Evidentials Haeger 2017	**Nominalized clauses** Howell 2018	**Valence change** Curtis 2018	**Adnominal possession** Nielsen & Bender 2018
Tense & aspect Poulson 2011	**Sentential negation** Crowgey 2012	**Information structure** Song 2014	**Adjectives & copulas** Trimble 2014
Morphotactics O'Hara 2008; Goodman 2013	**Person, number, gender** Drellishak 2009	**Agreement** Drellishak 2009	**Argument optionality** Saleem 2010
Coordination Drellishak & Bender 2005	**Polar questions** Bender & Flickinger 2005	**Lexicon** Drellishak & Bender 2005	**Case** Drellishak 2008
Core grammar Bender, Flickinger & Oepen 2002 (from the English Resource Grammar, Flickinger 2000)			

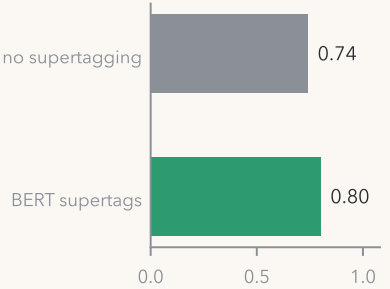
Libraries as listed in Zamaraeva, Curtis, Emerson, Fokkens, Goodman, Howell, Trimble & Bender (2022), *Journal of Language Modelling*.

Grammar, Faster



Machine learning helping a grammar: guessing each word's type first makes parsing faster, and more accurate.



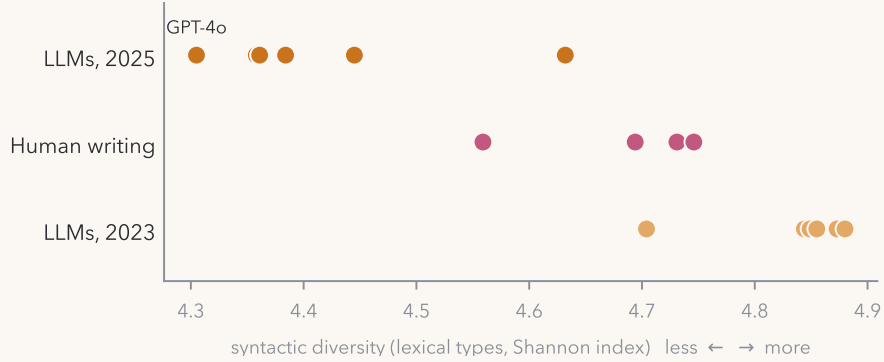


Averages over seven English test sets. Zamaraeva & Gómez-Rodríguez (2024), *Revisiting Supertagging for Faster HPSG Parsing*, EMNLP.

More Aligned, Less Diverse



The grammar as an instrument: newer, instruction-tuned models write with less syntactic variety than people do.

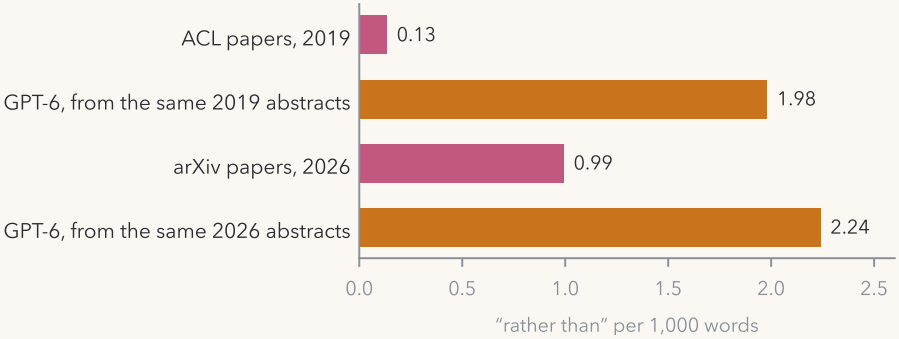


Each dot is a text source, parsed with the English Resource Grammar (Flickinger 2000). *More Aligned, Less Diverse? Analyzing the Grammar and Lexicon of Two Generations of LLMs* (ACL 2026).

Why Does This Text Annoy Me?



Why do so many recent papers grate on readers? Not just counting rather than: asking which uses annoy people, and why models produce them.

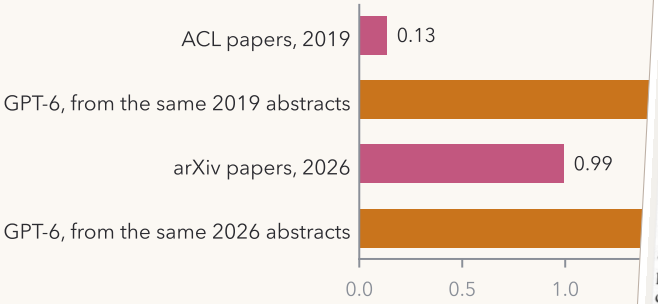


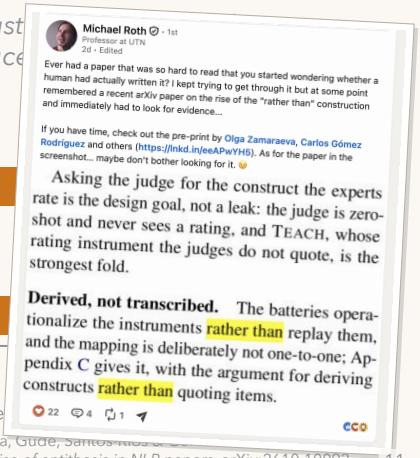
GPT-6 papers generated from the titles and abstracts of the human-written papers. Zamaraeva, Gude, Santos-Ríos & Gómez-Rodríguez (2026), *I would rather quit NLP than read another paper like this: The rise of antithesis in NLP papers*, arXiv:2610.10092.

Why Does This Text Annoy Me?



Why do so many recent papers grate on readers? Not just asking which uses annoy people, and why models produce





Michael Roth · 1st
Professor at UTN
2d · Edited

Ever had a paper that was so hard to read that you started wondering whether a human had actually written it? I kept trying to get through it but at some point remembered a recent arXiv paper on the rise of the "rather than" construction and immediately had to look for evidence...

If you have time, check out the pre-print by Olga Zamaraeva, Carlos Gómez Rodríguez and others (https://lnkd.in/eeAPwYH5). As for the paper in the screenshot... maybe don't bother looking for it.

Asking the judge for the construct the experts rate is the design goal, not a leak: the judge is zero-shot and never sees a rating, and TEACH, whose rating instrument the judges do not quote, is the strongest fold.

Derived, not transcribed. The batteries operationalize the instruments rather than replay them, and the mapping is deliberately not one-to-one; Appendix C gives it, with the argument for deriving constructs rather than quoting items.

22 4 1

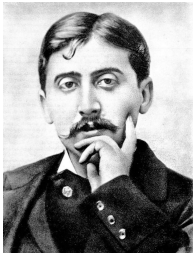
GPT-6 papers generated from the titles and abstracts of the human-written papers. Zamaraeva, Gude, Santos Rios [illegible]

I would rather quit NLP than read another paper like this: The rise of antithesis in NLP papers, arXiv:2610.10092.

Hard to Translate



Literary translation is slow. In Search of Lost Time took about seventy years to reach Russian readers in full, and two of its translators died before finishing.



Marcel Proust

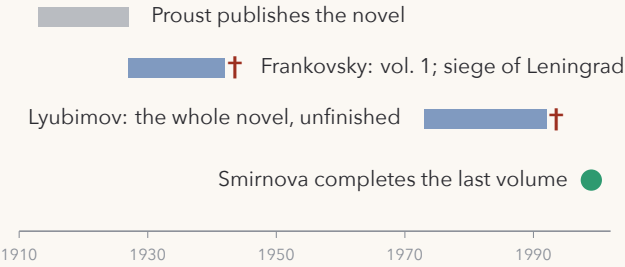
photo: Otto Wegener, c. 1895



Nikolai Lyubimov

translator, 1912–1992;

photo: perevodov.info



Dates from the publication history summarized at languagehat.com, “Proust in Russian”.

Harder to Find



In Proust, a unit of analysis is often not a sentence but a thread: a word, a taste, a melody that returns hundreds or thousands of pages later.

a taste, recalled: the madeleine … the uneven paving stones

a melody: Vinteuil's little phrase … the septet

1. Swann's Way	2. Young Girls in Flower	3. The Guermantes Way	4. Sodom and Gomorrah	5. The Prisoner	6. The Fugitive	7. Time Regained

In Search of Lost Time: seven volumes, about 3,000 pages

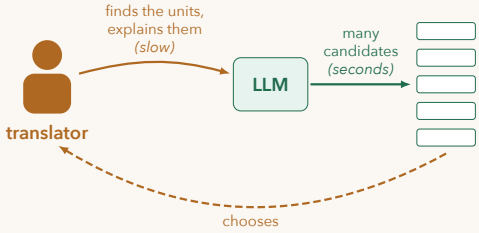
a word: *temps perdu* … *temps retrouvé*

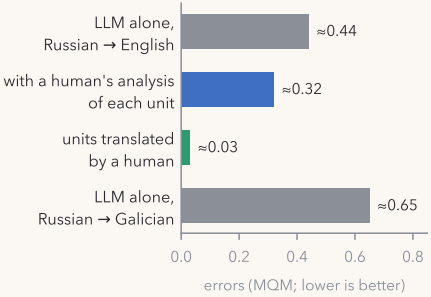
perdu is both "lost" and "wasted"; the Russian and English titles keep only "lost"

Harder to Find



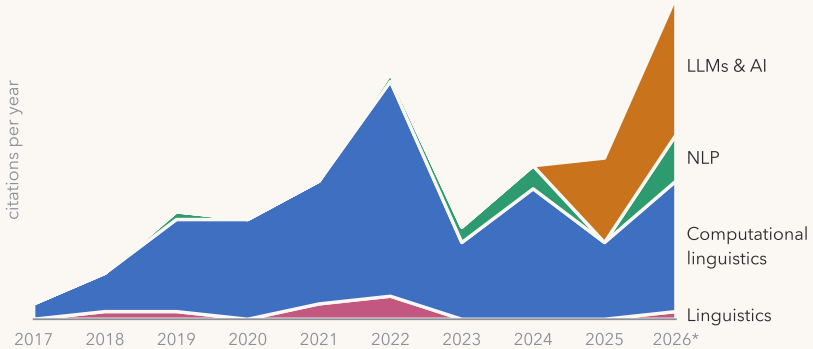
The bottleneck is not translating culturally loaded units, but finding them. Translator and LLM can split the work.





Subtitles of Soviet film comedies (here: *Ivan Vasilievich Changes Profession*); errors counted on the culturally loaded units. Adapted MQM error score, approximate values. Work in progress.

Footprints: citations per year, per field



Stacked chart: each band's thickness is that field's citations; the top edge is the total.
* 2026 to date. Source: Google Scholar.

What Puts Us on the Move?



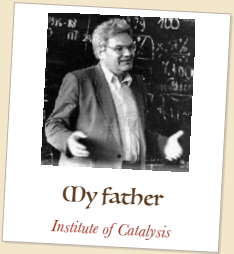
What Puts Us on the Move?



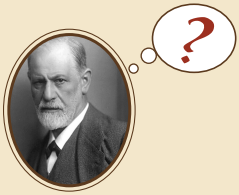
The Only Road?..







The Only Road?..







Akademgorodok

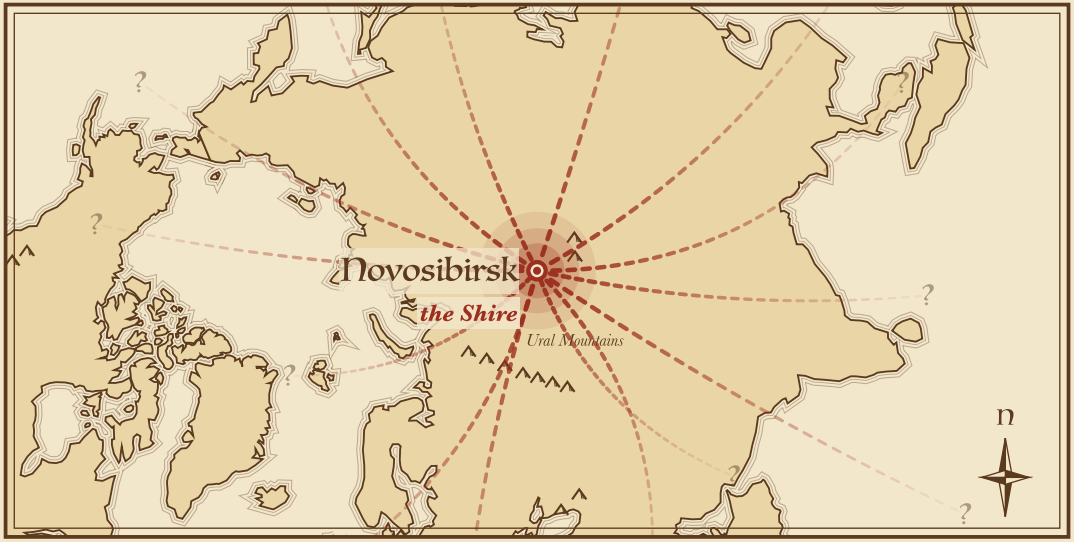
a town built for science



My father

Institute of Catalysis

Roads Not Taken



Animals, Fear, Curiosity, and Life



Lyudmila Trut, 1974

Photo: Svetlana Argutinskaya, CC BY-SA 4.0



A domesticated fox today

Photo: Kayfedewa, CC BY 3.0

Novosibirsk, since 1959

Foxes bred for one trait only: *less fear of people.*

Within a few generations: pets. And, incidentally, they started to look like dogs.

Less fear → more curiosity

Belyaev, D. K. (1979). Destabilizing selection as a factor in domestication. *Journal of Heredity* 70: 301--308.
Trut, L. N. (1999). Early canid domestication: The farm-fox experiment. *American Scientist* 87: 160--169.
Dugatkin, L. A., & Trut, L. (2017). *How to Tame a Fox (and Build a Dog)*. University of Chicago Press.

Fear or Curiosity?



Research is the only thing I could possibly do.

I can't leave my advisor's research line.

Why can't I work on literary MT?

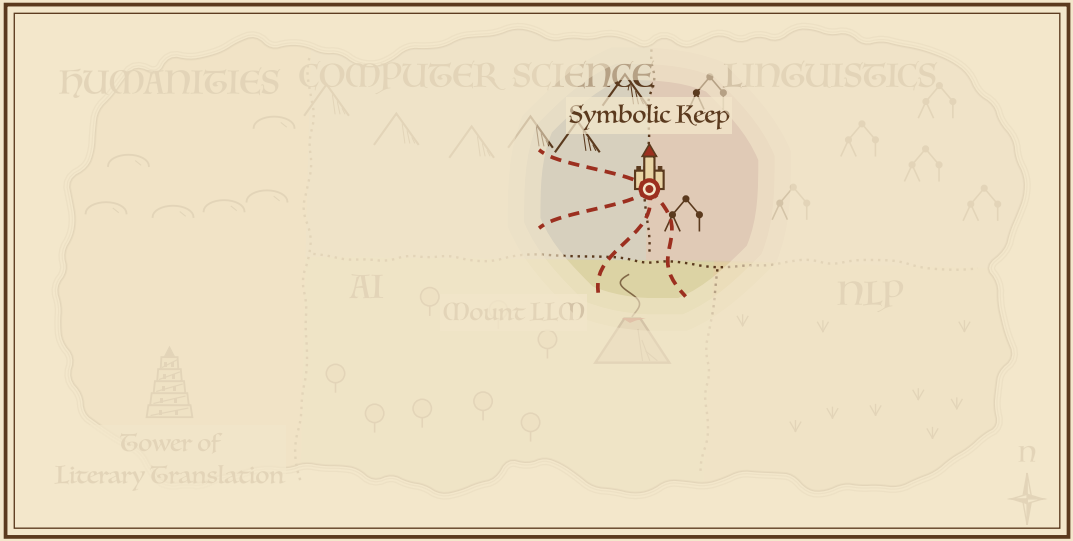
Can a machine translate something untranslatable?

This LLM-generated text annoys me. Why?

Fear of change is normal. But changes have to be attempted.

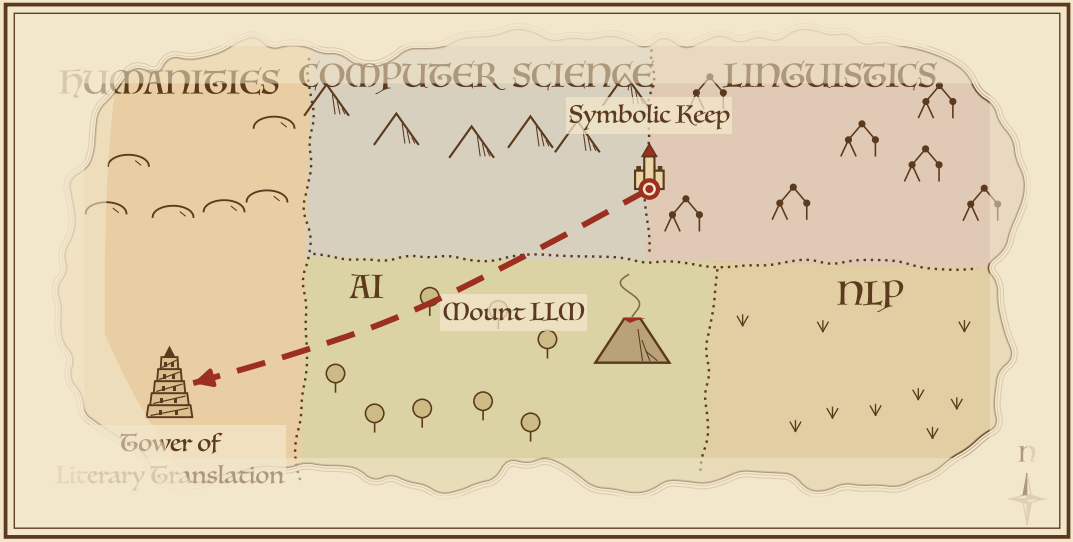
Range of Vision

with fear



Range of Vision

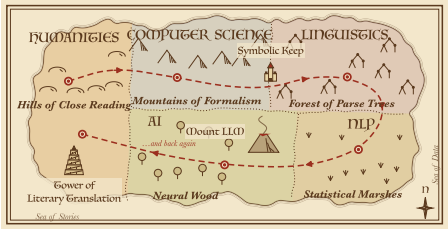
with less fear



Two Kinds of Change



geographical



intellectual

… and what each of them costs

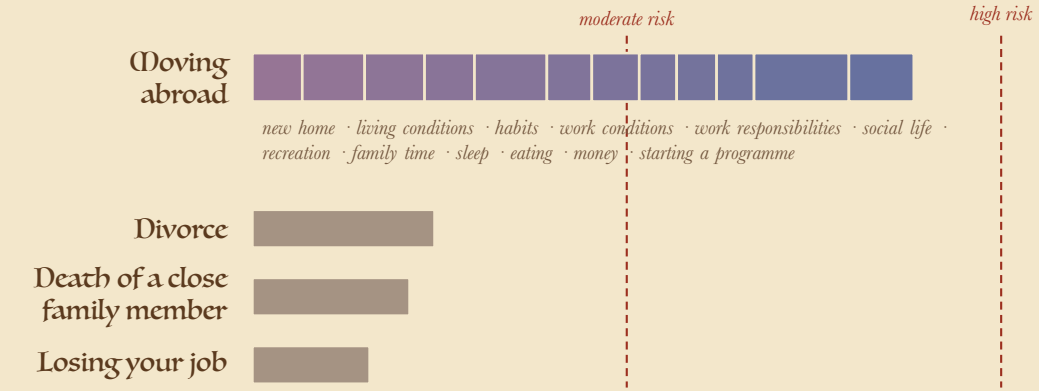
How Far from the Shire?

Home is behind, the world ahead, and there are many paths to tread. — J.R.R. Tolkien



Thin, Sort of Stretched

I feel thin, sort of stretched, like butter scraped over too much bread. — Bilbo Baggins



Item scores (life change units): Holmes, T. H., & Rahe, R. H. (1967). The Social Readjustment Rating Scale. *J. Psychosomatic Research* 11(2): 213--218.
Risk bands (150 and 300 units) as conventionally used with the scale; validation: Rahe, R. H., Mahan, J. L., & Arthur, R. J. (1970). Prediction of near-future health change from subjects' preceding life changes. *J. Psychosomatic Research* 14(4): 401--406.
``Moving abroad'' is the sum of the scale's items that a move typically involves (my selection). Quote: J. R. R. Tolkien, *The Fellowship of the Ring* (1954).

Visiting a Country vs. Living in It

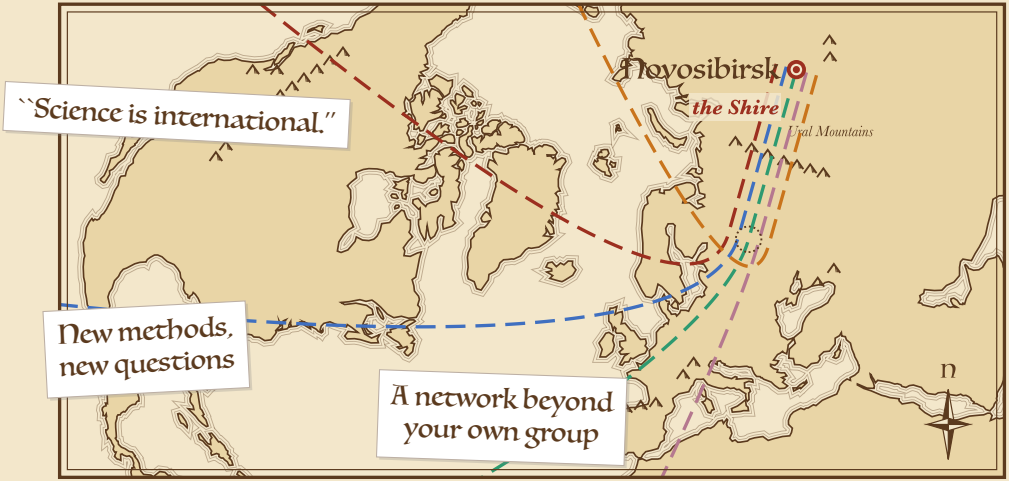
Language	*Get by in English*	***Spend tremendous energy learning to actually use it***
People	*Casual socializing*	***Meaningful, lasting relationships***
Dating	*??*	***??***

I bet you didn't expect me to talk about that here.

Not a question of time, but of energy; and energy is not a function of time.

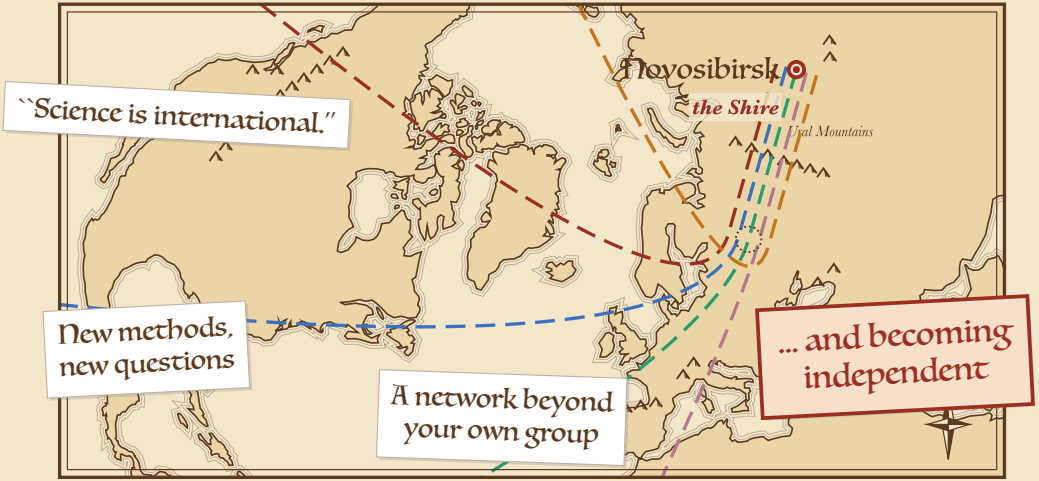
Why Go Abroad?

International experience is highly valued. Why?

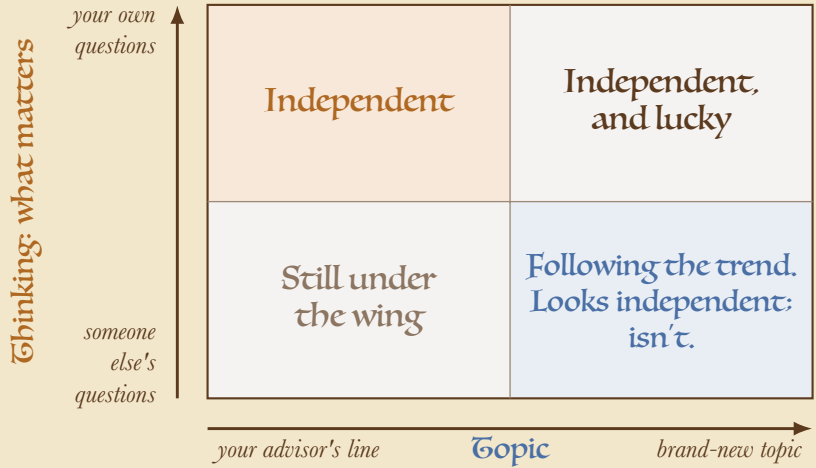


Why Go Abroad?

International experience is highly valued. Why?



Independent of What?



The two axes are sometimes confused by the very funding panels that ask for independence.

Leaving the Nest

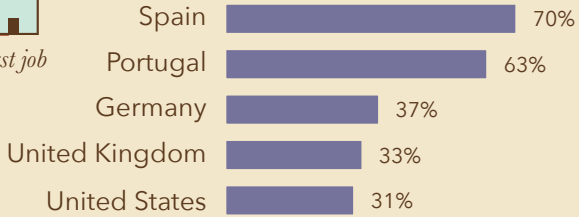
The American way



Staying home

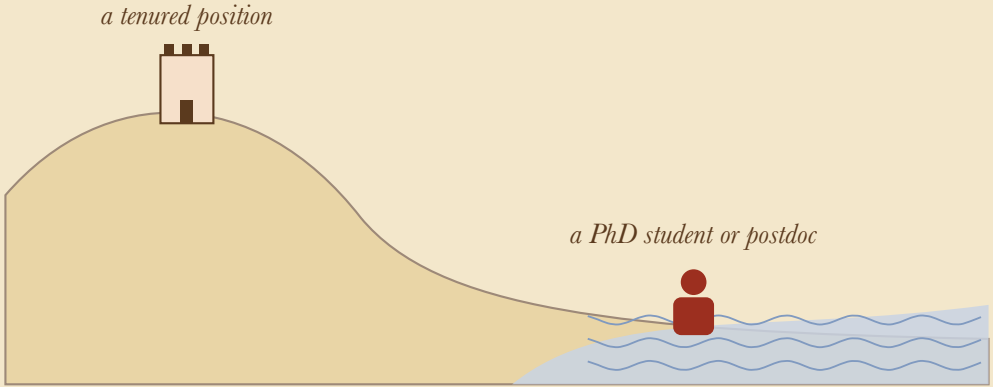


Researchers who started their career at the same university



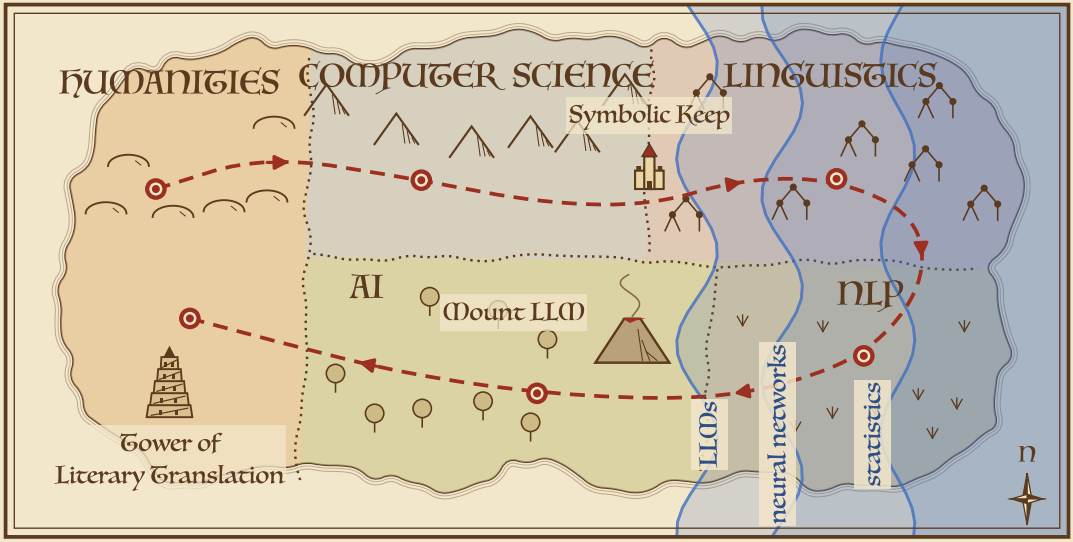
Leiden Ranking universities, all fields; public dataset, weighted by researchers.
Macháček et al. (2022), *Science and Public Policy* 49(1).

Who Feels the Tide?

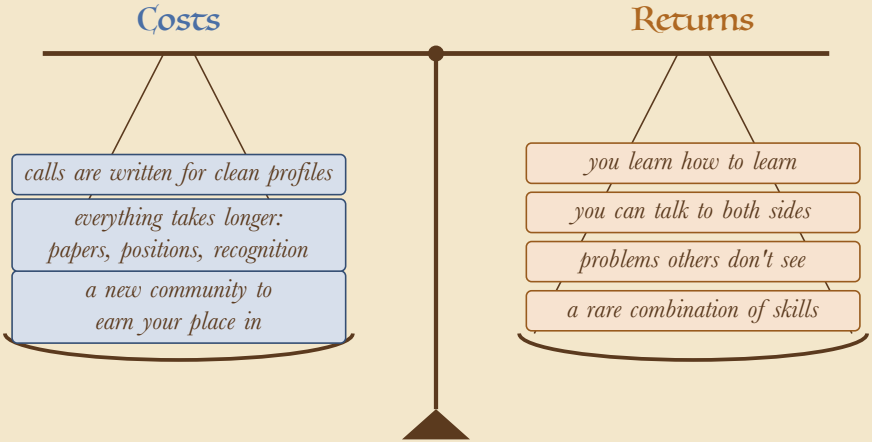


Sometimes becoming independent means breaking away from your PhD group's research line, especially when the field is changing.

The Rising Tide



What a Field Change Costs



How to Stop Worrying, Become Independent, and Change Fields When Necessary

Little by little, like domesticating foxes. (But maybe we don't want to be like those domesticated foxes after all.)



Fear — Curiosity

comfortable → *uncomfortable*

Wolves keep working on a puzzle alone; dogs turn to a human. Udell, M. A. R. (2015). When dogs look back. *Biology Letters* 11: 20150489.

For the Road

- Science is international, and has to be.
- Like many animals, we are moved by fear and by curiosity: in life, and in research careers.
- There is no right way. What matters is taking decisions to make turns, and working on maintaining calm in the process, which requires excellent mental health.
- Fear is normal. But try to notice when it is choosing for you.

All we have to decide is what to do with the time that is given us.

— Gandalf

Happy journey!

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